

QFT1 Formulas

Idan Stark

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1 A theory of two scalars in $5+1$ dimensions

1.1 Counter terms

We take the most general form of the Lagrangian with all renormalizable operators, i.e. all operators with dimension $\Delta \leq 3$. There are no symmetries, so this includes:

$$\begin{aligned} \mathcal{L} = \mathcal{L}_0 + \sum_i^2 \left(\frac{1}{2} \delta_{z_i} (\partial_\mu \phi_i)^2 - \frac{1}{2} \delta_{m_i} \phi_i^2 \right) - \\ - \delta_0 \phi_1 \phi_2 - \frac{1}{2} \delta_1 \phi_1^2 \phi_2 - \frac{1}{2} \delta_2 \phi_1 \phi_2^2 - \frac{1}{6} \delta_3 \phi_2^3 - \frac{1}{6} \delta_4 \phi_1^3 \end{aligned} \quad (1)$$

1.2 The correlation function $\langle \phi(x) \rangle$

The correlation function $\langle \phi(x) \rangle$, in one loop order, requires an odd number of ϕ_1 fields in the term, to contract with it. The only such term in the original Lagrangian is $\frac{\lambda_2}{2} \phi_1 \phi_2^2$. This is a tadpole diagram, with only one contraction option, giving:

$$\langle \tilde{\phi}(p) \rangle = \frac{-i\lambda_2 \delta^{(6)}(p)}{2} \frac{i}{p^2 - m_1^2 + i\epsilon} \int \frac{d^6 k}{(2\pi)^6} \frac{i}{k^2 - m_2^2 + i\epsilon} \quad (2)$$

This diverges quartically as $\frac{d^6 k}{k^2} \sim k^4$, so we will use dimensional regularization:

$$\int \frac{d^6 k}{(2\pi)^6} \frac{i}{k^2 - m_2^2 + i\epsilon} \rightarrow \int \frac{d^d k}{(2\pi)^d} \frac{i}{k^2 - m_2^2 + i\epsilon} \quad (3)$$

Moving to Euclidean space:

$$\int \frac{d^d k}{(2\pi)^d} \frac{i}{k^2 - m_2^2 + i\epsilon} = \int \frac{d^d k_E}{(2\pi)^d} \frac{1}{p_E^2 + m_2^2 + i\epsilon} \quad (4)$$

And using the known integral for $\Delta = m_2^2, n = 1$:

$$\langle \tilde{\phi}(p) \rangle = \frac{\lambda_2 \delta^{(d)}(p)}{2(4\pi)^{d/2}} \frac{1}{p^2 - m_1^2 + i\epsilon} \frac{\Gamma(1 - \frac{d}{2})}{\Gamma(1)} \Delta^{\frac{d}{2}-1} \quad (5)$$

$$\langle \tilde{\phi}(p) \rangle = \frac{1}{p^2 - m_1^2 + i\epsilon} \frac{\Gamma(1 - \frac{d}{2})}{2(4\pi)^{d/2}} \frac{1}{\Gamma(1)} m_2^{d-2} \lambda_2 \delta^{(d)}(p) \quad (6)$$

Since

$$\left[\frac{1}{p^2 - m_1^2 + i\epsilon} \right] = -2 \quad [m_2] = 1 \quad [\lambda_2] = 3 - \frac{d}{2} \quad [\delta^{(d)}(p)] = -d \quad (7)$$

The dimension of the correlation function is:

$$[\langle \tilde{\phi}(p) \rangle] = -2 + (d-2)[m_2] + [\lambda_2] - d = -\frac{d+2}{2} \quad (8)$$

Which, after performing the Fourier transform back to $\phi(x)$, gives (with the measure of the fourier integral having mass dimension d), exactly the field mass dimension $\frac{d-2}{2}$.

As we can see, this result is linear in λ_2 , so a counter term of the same form $\phi_1 \phi_2^2$ will give similar result. Setting $\delta_2 = -\lambda_2$ will give the opposite contribution, making the correlation function vanish.

1.3 The 1PI propagators

1.3.1 $\langle \phi_1(p) \phi_1(-p) \rangle$

In this case there are two contributing diagrams:



They give:

$$\begin{aligned} -i\Sigma_{11}(p) &= \frac{(-i\lambda_2)^2}{2} \int \frac{d^6 k}{(2\pi)^k} \frac{i}{k^2 - m_2^2 + i\epsilon} \frac{i}{(p-k)^2 - m_2^2 + i\epsilon} + \\ &+ \frac{(-i\lambda_1)^2}{2} \int \frac{d^6 k}{(2\pi)^k} \frac{i}{k^2 - m_1^2 + i\epsilon} \frac{i}{(p-k)^2 - m_2^2 + i\epsilon} \end{aligned} \quad (9)$$

Moving to d dimensions and using the given integral:

$$-i\Sigma_{11}(p) = i \frac{\Gamma(2 - \frac{d}{2})}{2(2\pi)^{d/2}} \int_0^1 dx \left[\frac{\lambda_2^2}{(m_2^2 - x(1-x)p^2)^{2-\frac{d}{2}}} + \frac{\lambda_1^2}{(xm_1^2 + (1-x)m_2^2 - x(1-x)p^2)^{2-\frac{d}{2}}} \right] \quad (10)$$

1.3.2 $\langle \phi_1(p)\phi_2(-p) \rangle$

This case is very similar, with again two diagrams:



This is identical to the first case, only with different coefficients matching the vertices, and we need to make sure the internal propagators have the correct mass:

$$-i\Sigma_{12}(p) = i \frac{\Gamma\left(2 - \frac{d}{2}\right)}{2(2\pi)^{d/2}} \int_0^1 dx \left[\frac{\lambda_2 \lambda_3}{\left(m_2^2 - x(1-x)p^2\right)^{2-\frac{d}{2}}} + \frac{\lambda_1 \lambda_2}{\left(xm_1^2 + (1-x)m_2^2 - x(1-x)p^2\right)^{2-\frac{d}{2}}} \right] \quad (11)$$

1.4 Imaginary part of Σ_{11}

In order to have an imaginary part, there should be physical intermediate states. In the correlation function $\langle \phi_1(p)\phi_1(-p) \rangle$, this means there are enough momentum to create either two ϕ_2 particles, or a ϕ_1 and a ϕ_2 particle:

$$p^2 \geq \min(4m_2^2, (m_1 + m_2)^2) = \begin{cases} 4m_2^2 & m_1 < m_2 \\ (m_1 + m_2)^2 & m_2 < m_1 \end{cases} \quad (12)$$

1.5 Counter terms

The 2 point correlation function gets contribution from the kinetic and mass counter terms:

$$\Sigma_{11}(p) = \frac{\Gamma\left(2 - \frac{d}{2}\right)}{2(2\pi)^{d/2}} \int_0^1 dx \left[\frac{\lambda_2^2}{\left(m_2^2 - x(1-x)p^2\right)^{2-\frac{d}{2}}} + \frac{\lambda_1^2}{\left(xm_1^2 + (1-x)m_2^2 - x(1-x)p^2\right)^{2-\frac{d}{2}}} \right] + \delta_{z_1} p^2 - \delta_{m_1} \quad (13)$$

Deriving by p^2 gives:

$$\begin{aligned} \frac{\partial}{\partial p^2} \Sigma_{11}(p) &= \delta_{z_1} + \\ &+ \left(2 - \frac{d}{2}\right) \frac{\Gamma\left(2 - \frac{d}{2}\right)}{2(2\pi)^{d/2}} \int_0^1 dx \left[\frac{\lambda_2^2 x(1-x)}{\left(m_2^2 - x(1-x)p^2\right)^{3-\frac{d}{2}}} + \frac{\lambda_1^2 x(1-x)}{\left(xm_1^2 + (1-x)m_2^2 - x(1-x)p^2\right)^{3-\frac{d}{2}}} \right] \end{aligned} \quad (14)$$

Requiring the derivative to be zero for $p^2 = M^2$ gives:

$$\delta_{z_1} = \left(\frac{d}{2} - 2\right) \frac{\Gamma\left(2 - \frac{d}{2}\right)}{2(2\pi)^{d/2}} \int_0^1 dx \left[\frac{\lambda_2^2 x(1-x)}{(m_2^2 - x(1-x)M^2)^{3-\frac{d}{2}}} + \frac{\lambda_1^2 x(1-x)}{(xm_1^2 + (1-x)m_2^2 - x(1-x)M^2)^{3-\frac{d}{2}}} \right] \quad (15)$$

Similarly from the Σ_{12} 1PI, we get:

$$\delta_{z_2} = \left(\frac{d}{2} - 2\right) \frac{\Gamma\left(2 - \frac{d}{2}\right)}{2(2\pi)^{d/2}} \int_0^1 dx \left[\frac{\lambda_2 \lambda_3 x(1-x)}{(m_2^2 - x(1-x)M^2)^{3-\frac{d}{2}}} + \frac{\lambda_1 \lambda_2 x(1-x)}{(xm_1^2 + (1-x)m_2^2 - x(1-x)M^2)^{3-\frac{d}{2}}} \right] \quad (16)$$

1.6 The physical masses of the particles

1.6.1 Stability of the particles

A prticle in this theory is unstable only if it can decay into a pair of the other particle. Therefore, for $\tilde{m} < 2m$ and $m < 2\tilde{m}$, the particles are stable. This gives:

$$\frac{1}{2} < \frac{m}{\tilde{m}} < 2 \quad (17)$$

1.6.2 The poles and imaginary part of $\langle \phi_1 \phi_1 \rangle$

Since the particles are a generic mix of the two fields, inverting the transformation means that each field is a generic combination of the two particle fields. In other words, the propagator of each field will have poles for both physical masses $m, \tilde{m}$. Additionally, the $\langle \phi_1 \phi_1 \rangle$ propagator will have an imaginary part if it has a physical intermediate state, which means that it will create two physical particles. We take the one with the smaller mass:

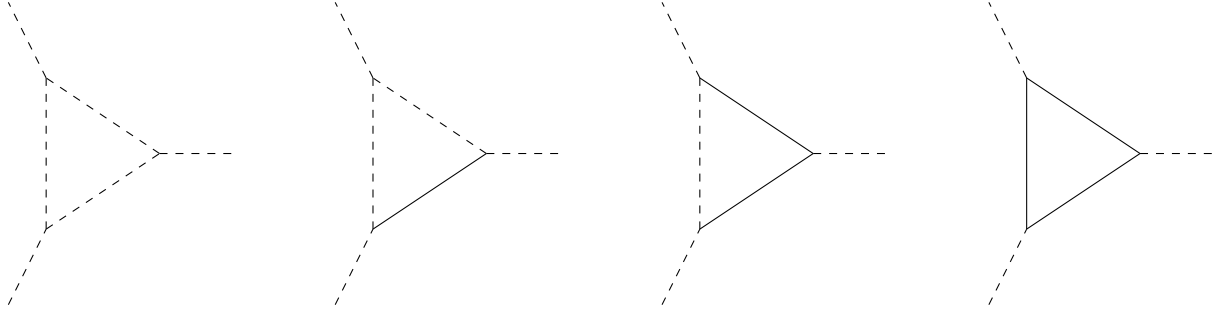
$$p^2 \geq 4 \min(m, \tilde{m})^2 = \begin{cases} 4m^2 & m < \tilde{m} \\ 4\tilde{m}^2 & \tilde{m} < m \end{cases} \quad (18)$$

1.7 The beta functions of λ_1 , λ_2 , and λ_3

We saw in the tutorial that for the ϕ^3 theory there is a single diagram contributing to the vertex correction in order λ^3 , which gave us, for scales much larger then the masses:

$$-\frac{\lambda^3}{(4\pi)^3} \quad (19)$$

In the $\langle\phi_2\phi_2\phi_2\rangle$ case of this theory, we have eight different diagrams:

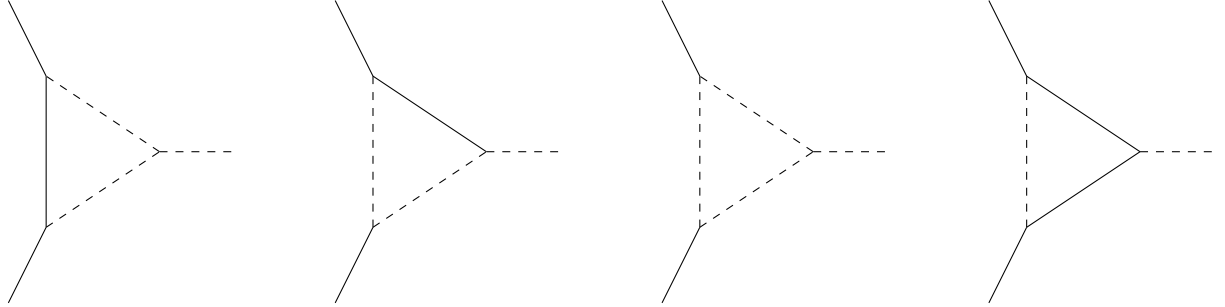


Where the center two diagrams have each three different positions for the internal ϕ_1 propagator. From left to right, they give the contributions:

$$\beta(\lambda_3) = -2 \cdot \frac{1}{2} \frac{\lambda_3^3}{(2\pi)^3} - 2 \cdot 3 \cdot \frac{1}{9} \frac{\lambda_2^2 \lambda_3}{(2\pi)^3} - 2 \cdot 3 \cdot \frac{2}{27} \frac{\lambda_1 \lambda_2^2}{(2\pi)^3} - 2 \cdot \frac{1}{27} \frac{\lambda_1^3}{(2\pi)^3} \quad (20)$$

$$\beta(\lambda_3) = -\frac{1}{(2\pi)^3} \left[\lambda_3^3 + \frac{2}{3} \lambda_2^2 \lambda_3 + \frac{4}{9} \lambda_1 \lambda_2^2 + \frac{2}{27} \lambda_1^3 \right] \quad (21)$$

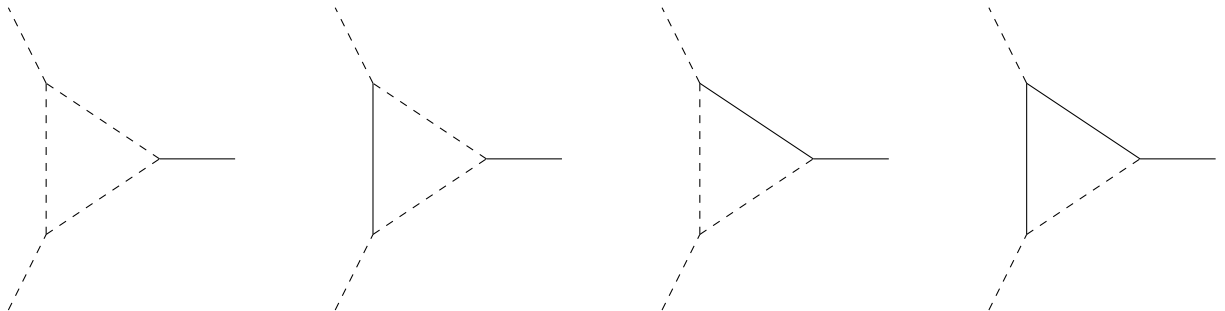
Similarly, each of the other vertices has four diagrams contributing to its correction. For the $\langle\phi_1\phi_1\phi_2\rangle$ correlation function, the diagrams are:



Where the second diagram has two permutations of the internal propagators. This gives the beta function:

$$\beta(\lambda_1) = -\frac{1}{(2\pi)^3} \left[\frac{8}{9} \lambda_1^2 \lambda_3 + \frac{8}{27} \lambda_1 \lambda_2^2 + \frac{4}{27} \lambda_1^3 + \frac{2}{9} \lambda_2^2 \lambda_3^2 \right] \quad (22)$$

And for $\langle\phi_2\phi_2\phi_1\rangle$, the diagrams are:



Where the third and fourth diagrams have two permutations of the internal propagators. This gives the beta function:

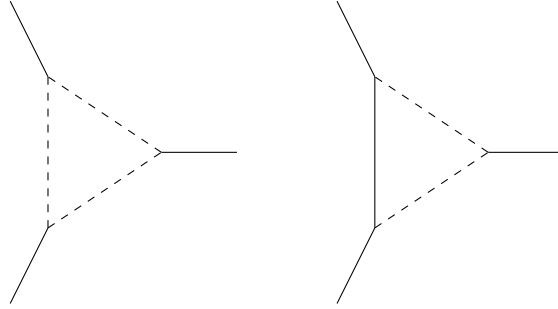
$$\beta(\lambda_2) = -\frac{1}{(2\pi)^3} \left[\frac{2}{3}\lambda_2^2\lambda_3 + \frac{4}{27}\lambda_2^3 + \frac{2}{9}\lambda_1\lambda_2\lambda_3 + \frac{4}{27}\lambda_1^2\lambda_2 \right] \quad (23)$$

1.8 Linear terms in the coupling of in the beta functions

While in the perturbative sense, all terms are of order λ^3 , there are terms in the beta functions that are linear to each of the λ_i .

1.9 The remaining coupling

The last dimensionless coupling would be the coupling for the ϕ_1^3 operator, named λ_4 by our convention. the one loop contribution to the correction has just two diagrams:



Where the second diagram again has three permutation of the internal propagators. This gives the beta function:

$$\beta(\lambda_4) = -\frac{1}{(2\pi)^3} \left[\frac{2}{27}\lambda_2^3 + \frac{4}{9}\lambda_1^2\lambda_2 \right] \quad (24)$$

1.10 The $\lambda_2 = 0$ case

1.10.1 The symmetry

This gives the Lagrangian

$$L = \sum_{i=1}^2 \left(\frac{1}{2} (\partial_m \phi_i)^2 - \frac{1}{2} m_i^2 \phi_i^2 \right) - \frac{1}{2} \lambda_1 \phi_1^2 \phi_2 - \frac{1}{6} \lambda_3 \phi_2^3 \quad (25)$$

This theory has a symmetry:

$$\phi_1 \rightarrow -\phi_1 \quad (26)$$

1.10.2 The consistancy of the results from previous sections

We can see that if $\lambda_2 = 0$ for some scale M , then the beta function for both λ_2 and λ_4 is zero, and thus these terms will not be generated by the RG flow, so these results are consistant with the symmetry.

1.10.3 The change is reulsts to f (1.6)

Since the fields do not mix (we saw that in the first term I can copy the results here, but basically $\langle \phi_1 \phi_2 \rangle = -\langle \phi_1 \phi_2 \rangle$ from the path integral with integration variable change), then the masses aer m_1 and m_2 .

The result of the first section is the same: for a particle to be stable it cannot decay into the other kind of particle through the interactions, meaning that for both to be stable we require:

$$\frac{1}{2} < \frac{m_1}{m_2} < 2 \quad (27)$$

For the second part, now the poles of the Σ_{ii} 1PI will only be for m_1 , and it will have an imaginary part by the same condition:

$$p^2 > 4\min(m_1, m_2)^2 = \begin{cases} 4m_1^2 & m_1 < m_2 \\ 4m_2^2 & m_2 < m_1 \end{cases} \quad (28)$$

2 Photon-photon scattering in scalar QED

2.1 The effective photon action

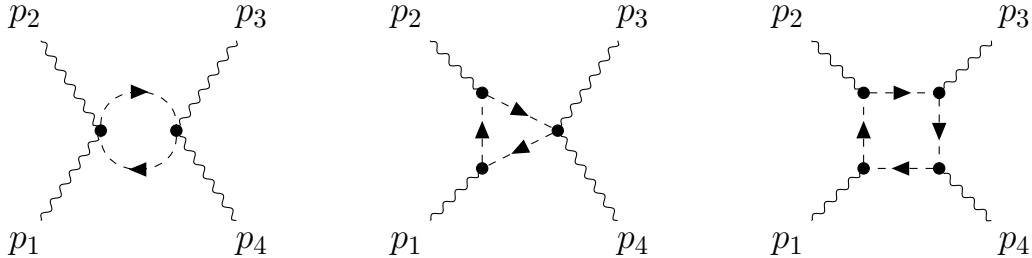
The most general term that depends on the photon alone that is gauge invariant and Lorentz invariant is $F_{\mu\nu}$. This term is linear in A_μ , and so to adhere to $A_\mu \rightarrow -A_\mu$ symmetry, we must take even powers of it. This means that after the kinetic term, the term with the lowest dimension is the quartic term:

$$\mathcal{L}_A = -\frac{1}{4}F_{\mu\nu}^2 - \frac{g}{16}F_{\mu\nu}^4 \quad (29)$$

2.2 The first order contributing diagrams

We work in Feynman gauge, but since it only contributes to the photon propagator, it wouldn't change the value of this result.

Since there are two types of vertices contributing to this process at one-loop order (from the operators $A^2\phi^2$ and $A\phi^2$), there are three families of diagrams: diagrams with all photons interacting with the scalar fields alone, diagrams where all photons interact with the scalar fields in pairs, and diagrams where two photons interact in a pair and two interact alone:



Each of these diagrams has a different number of permutations over external momenta. From left to right, we name them $i\mathcal{M}_{1,2,3}(p_i, \epsilon^i)$. For $i\mathcal{M}_1$ we get the three options of the Mandelstam channels:

$$i\mathcal{M}_I = i\mathcal{M}_1 + i\mathcal{M}_1(p_2 \leftrightarrow p_4) + i\mathcal{M}_1(p_2 \leftrightarrow p_3) \quad (30)$$

For $i\mathcal{M}_2$ we get all possible permutations of a circle with a start position (just regular order, $4!$), but with two of the photons being at the same place, dividing

it by 2 to give 12 options:

$$\begin{aligned}
i\mathcal{M}_{II} = & i\mathcal{M}_2 + i\mathcal{M}_2(p_2 \leftrightarrow p_3) + i\mathcal{M}_2(p_2 \leftrightarrow p_4) + \\
& + i\mathcal{M}_2(p_1 \leftrightarrow p_2) + i\mathcal{M}_2(p_1 \leftrightarrow p_3) + i\mathcal{M}_2(p_1 \leftrightarrow p_4) + \\
& + i\mathcal{M}_2(p_2 \leftrightarrow p_4, p_1 \leftrightarrow p_3) + i\mathcal{M}_2(p_2 \leftrightarrow p_3, p_1 \leftrightarrow p_4) + \\
& + i\mathcal{M}_2(p_1 \rightarrow p_2 \rightarrow p_3 \rightarrow p_1) + i\mathcal{M}_2(p_1 \rightarrow p_3 \rightarrow p_2 \rightarrow p_1) + \\
& + i\mathcal{M}_2(p_1 \rightarrow p_2 \rightarrow p_4 \rightarrow p_1) + i\mathcal{M}_2(p_1 \rightarrow p_4 \rightarrow p_2 \rightarrow p_1)
\end{aligned} \tag{31}$$

For $i\mathcal{M}_3$ we get all permutations of orders in a circle, a total of six:

$$\begin{aligned}
i\mathcal{M}_{III} = & i\mathcal{M}_3 + i\mathcal{M}_3(p_2 \leftrightarrow p_4) + i\mathcal{M}_3(p_2 \leftrightarrow p_4) + \\
& + i\mathcal{M}_3(p_3 \leftrightarrow p_4) + i\mathcal{M}_3(p_2 \rightarrow p_3 \rightarrow p_4 \rightarrow p_2) + i\mathcal{M}_3(p_2 \rightarrow p_4 \rightarrow p_3 \rightarrow p_2)
\end{aligned} \tag{32}$$

2.3 Calculating in the special case $p_i = 0$

The first diagram gives:

$$i\mathcal{M}_1 = \epsilon_\mu^1 \epsilon_\nu^2 (2ie^2)^2 \eta_{\mu\nu} \eta_{\sigma\rho} \int \frac{d^d k}{(2\pi)^d} \frac{i}{k^2 - M^2 + i\epsilon} \frac{i}{(k + p_1 + p_2)^2 - M^2 + i\epsilon} \epsilon_\sigma^{3*} \epsilon_\rho^{4*} \tag{33}$$

But, using $p_1 = p_2 = 0$, this becomes:

$$i\mathcal{M}_1 = \epsilon_\mu^1 \epsilon_\nu^2 (2ie^2)^2 \eta_{\mu\nu} \eta_{\sigma\rho} \int \frac{d^d k}{(2\pi)^d} \left(\frac{i}{k^2 - M^2 + i\epsilon} \right)^2 \epsilon_\sigma^{3*} \epsilon_\rho^{4*} \tag{34}$$

Moving to Eucliden coordinates and performing the integral using the formula given:

$$i\mathcal{M}_1 = i4e^4 \frac{\Gamma(2 - \frac{d}{2})}{(2\pi)^{d/2} \Gamma(2)} (M^2)^{\frac{d}{2}-2} \epsilon_\mu^1 \epsilon_\mu^2 \epsilon_\nu^{3*} \epsilon_\nu^{4*} \tag{35}$$

There are two additional permutations, their sum can be written as:

$$i\mathcal{M}_I = i4e^4 \frac{\Gamma(2 - \frac{d}{2})}{(2\pi)^{d/2}} (M^2)^{\frac{d}{2}-2} \varepsilon \tag{36}$$

Where we defined:

$$\varepsilon = (\epsilon_\mu^1 \epsilon_\mu^2 \epsilon_\nu^{3*} \epsilon_\nu^{4*} + \epsilon_\mu^1 \epsilon_\nu^2 \epsilon_\mu^{3*} \epsilon_\nu^{4*} + \epsilon_\mu^1 \epsilon_\nu^2 \epsilon_\nu^{3*} \epsilon_\mu^{4*}) \tag{37}$$

The second diagram gives (after taking the momenta to be zero again, causing the entire internal loop to have the same momentum):

$$\begin{aligned}
i\mathcal{M}_2 &= \epsilon_\mu^1 \epsilon_\nu^2 (-ie)^2 (2ie^2) \eta_{\sigma\rho} \int \frac{d^d k}{(2\pi)^d} (2k)^\mu (2k)^\nu \left(\frac{i}{k^2 - M^2 + i\epsilon} \right)^3 \epsilon_\sigma^{3*} \epsilon_\rho^{4*} = \\
&= -8e^4 \int \frac{d^d k}{(2\pi)^d} k^\mu k^\nu \left(\frac{1}{k^2 - M^2 + i\epsilon} \right)^3 \epsilon_\mu^1 \epsilon_\nu^2 \epsilon_\sigma^{3*} \epsilon_\sigma^{4*} = \\
&= -\frac{8}{d} e^4 \int \frac{d^d k}{(2\pi)^d} \frac{k^2}{(k^2 - M^2 + i\epsilon)^3} \epsilon_\mu^1 \epsilon_\mu^2 \epsilon_\nu^{3*} \epsilon_\nu^{4*}
\end{aligned} \tag{38}$$

Where in the last line we used:

$$\eta_{\mu\nu} \int d^d k k^2 f(k^2) = C \int d^d k k_\mu k_\mu f(k^2) \tag{39}$$

$$\eta^{\mu\nu} \eta_{\mu\nu} \int d^d k k^2 f(k^2) = C \eta^{\mu\nu} \int d^d k k_\mu k_\mu f(k^2) \tag{40}$$

$$d \int d^d k k^2 f(k^2) = C \int d^d k k^2 f(k^2) \tag{41}$$

$$d = C \tag{42}$$

$$\eta_{\mu\nu} \int d^d k k^2 f(k^2) = d \int d^d k k_\mu k_\mu f(k^2) \tag{43}$$

Moving to Euclidean space and solving the integral gives:

$$i\mathcal{M}_2 = i \frac{8}{d} e^4 \frac{d}{2} \frac{\Gamma\left(3 - \frac{d}{2} - 1\right)}{(2\pi)^{d/2} \Gamma(3)} (M^2)^{\frac{d}{2} + 1 - 3} \epsilon_\mu^1 \epsilon_\mu^2 \epsilon_\nu^{3*} \epsilon_\nu^{4*} \tag{44}$$

$$i\mathcal{M}_2 = i 2e^4 \frac{\Gamma\left(2 - \frac{d}{2}\right)}{(2\pi)^{d/2}} (M^2)^{\frac{d}{2} - 2} \epsilon_\mu^1 \epsilon_\mu^2 \epsilon_\nu^{3*} \epsilon_\nu^{4*} \tag{45}$$

There are twelve combinations this time. Four of them will give off each of the pairings of polarizations, so we get a factor of three:

$$i\mathcal{M}_{II} = i 16e^4 \frac{\Gamma\left(2 - \frac{d}{2}\right)}{(2\pi)^{d/2}} (M^2)^{\frac{d}{2} - 2} \epsilon \tag{46}$$

Finally, we perform the third diagram:

$$\begin{aligned}
iM_3 &= \epsilon_\mu^1 \epsilon_\nu^2 (-ie)^4 \int \frac{d^d k}{(2\pi)^d} (2k)^\mu (2k)^\nu (2k)^\sigma (2k)^\rho \left(\frac{i}{k^2 - M^2 + i\epsilon} \right)^4 \epsilon_\sigma^{3*} \epsilon_\rho^{4*} = \\
&= 16e^4 \int \frac{d^d k}{(2\pi)^d} k^\mu k^\nu k^\sigma k^\rho \frac{1}{(k^2 - M^2 + i\epsilon)^4} \epsilon_\mu^1 \epsilon_\nu^2 \epsilon_\sigma^{3*} \epsilon_\rho^{4*}
\end{aligned} \tag{47}$$

Performing the same trick:

$$C \int \frac{d^d k}{(2\pi)^d} k^\mu k^\nu k^\sigma k^\rho f(k^2) = (\eta_{\mu\nu}\eta_{\rho\sigma} + \eta_{\mu\rho}\eta_{\nu\sigma} + \eta_{\mu\sigma}\eta_{\rho\nu}) \int \frac{d^d k}{(2\pi)^d} f(k^2) \quad (48)$$

Multiplying both sides by $\eta^{\mu\nu}\eta^{\rho\sigma}$, we get:

$$C \int \frac{d^d k}{(2\pi)^d} (k^2)^2 f(k^2) = (d^2 + 2d) \int \frac{d^d k}{(2\pi)^d} f(k^2) \quad (49)$$

$$C = d(d+2) \quad (50)$$

This gives:

$$iM_3 = \frac{16}{d(d+2)} e^4 \int \frac{d^d k}{(2\pi)^d} \frac{(k^2)^2}{(k^2 - M^2 + i\epsilon)^4} \epsilon \quad (51)$$

Moving the Euclidean space and performing the integral using the formula:

$$iM_3 = i \frac{16}{d(d+2)} e^4 \frac{d(d+2)}{4} \frac{\Gamma(4 - \frac{d}{2} - 2)}{(2\pi)^{d/2} \Gamma(4)} (M^2)^{\frac{d}{2} + 2 - 4} \epsilon \quad (52)$$

$$iM_3 = i4e^4 \frac{\Gamma(2 - \frac{d}{2})}{(2\pi)^{d/2} \Gamma(4)} (M^2)^{\frac{d}{2} - 2} \epsilon \quad (53)$$

Adding all six option together cancels the $\Gamma(4) = 6$ in the denominator, giving:

$$i\mathcal{M}_{III} = i4e^4 \frac{\Gamma(2 - \frac{d}{2})}{(2\pi)^{d/2}} (M^2)^{\frac{d}{2} - 2} \epsilon \quad (54)$$

Adding all three of them together gives:

$$i\mathcal{M} = i24e^4 \frac{\Gamma(2 - \frac{d}{2})}{(2\pi)^{d/2}} (M^2)^{\frac{d}{2} - 2} \epsilon \quad (55)$$

2.4 The effective action at low momenta

2.4.1 The contribution of the term from 2.1 (a)

The term we wrote $-\frac{g}{16} F_{\mu\nu}^4$ contributes directly to the four photon scattering amplitude at tree level. However, since the term has derivatives, the contribution of the vertex is zero for $p_i = 0$ (it has a multiplicative factor $((p^\mu + p^\nu)^2)^2$).

2.5 The mismatch

It seems that the result from our calculation is not matched to the generic form of an effective action in low momenta for a photon-only effective theory, since the most general terms we wrote are still related to derivatives and thus cancel out. This is a result of taking the momenta to zero, since for photons, this is not a physical limit.

2.6 Imaginary part of the $2 \rightarrow 2$ scattering amplitude

For any non zero momenta, since for these values there exists some q such that $(p_1 + p_2)^2 \geq 4q^2$ so a photon pair with momenta q and $-q$ can be created and annihilated as an intermediate physical state.

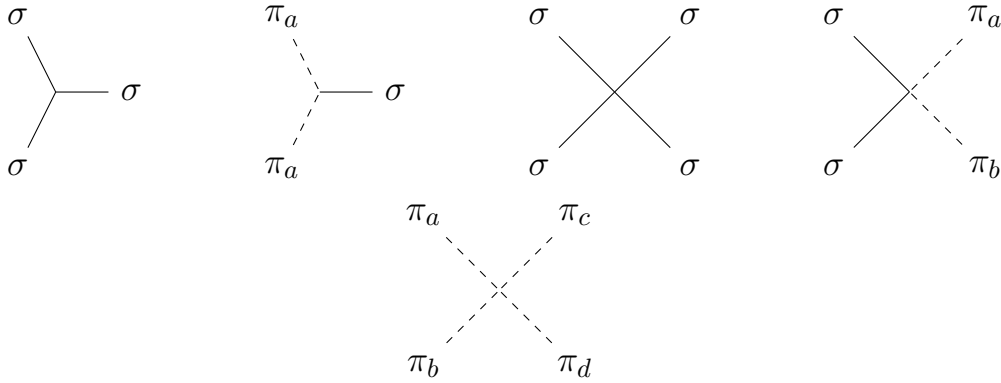
3 Spontaneous symmetry breaking

3.1 Feynman rules for the effective theory

These are the Feynman rules for scalar fields theory. So, for every diagram:

- Each σ propagator gives $\frac{i}{p^2 - 2\mu^2 + i\epsilon}$.
- Each π_a propagator gives $\frac{i}{p^2 + i\epsilon}$, since the π_a fields don't have a mass term.

There are then five types of vertices (σ is a solid line and π_a are dashed lines):



These give, from left to right. The operators in the Lagrangians don't all have the full combinatorial prefactor, so some get an additional prefactor here:

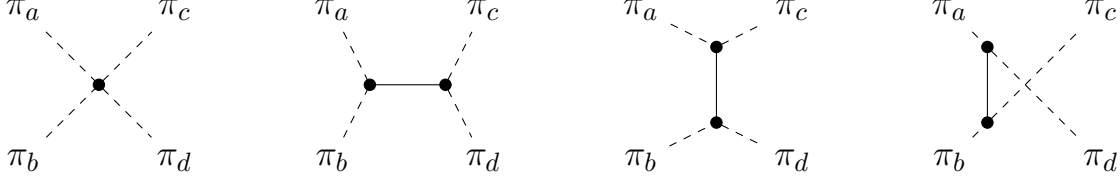
- $\sigma^3 \rightarrow -i6 \sqrt{\lambda} \mu$
- $\pi_a \pi_b \sigma \rightarrow -i2 \sqrt{\lambda} \mu \delta_{ab}$
- $\sigma^4 \rightarrow -i6 \lambda$
- $\pi_a \pi_b \sigma^2 \rightarrow -i2 \lambda \delta_{ab}$
- $\pi_a \pi_b \pi_c \pi_d \rightarrow -i6 \lambda (\delta_{ab} \delta_{cd} + \delta_{ac} \delta_{bd} + \delta_{ad} \delta_{bc})$

And, of course:

- Add a $(2\pi)^d \delta(\sum p_i)$ for each vertex
- Divide by symmetry factor
- Integrate over all undertermined momenta

3.2 Scattering of two Nambu-Goldstone bosons

Using the above vertices, we can see that there is a single vertex that has a tree-level contribution, which is the π_a^4 vertex, and another vertex that can give contribution at the same order in λ , but with a loop, which is the $\sigma\pi_a^2$ vertex. This second vertex gives the three Mandelstam channels:



This will give:

$$i\mathcal{M}_{abcd} = -i6\lambda(\delta_{ab}\delta_{cd} + \delta_{ac}\delta_{bd} + \delta_{ad}\delta_{bc}) - 4\lambda\mu^2 \left[\delta_{ab}\delta_{cd} \frac{i}{s - 2\mu^2 + i\epsilon} + \delta_{ac}\delta_{bd} \frac{i}{t - 2\mu^2 + i\epsilon} + \delta_{ad}\delta_{bc} \frac{i}{u - 2\mu^2 + i\epsilon} \right] \quad (56)$$

Now, for $p_i^2 = 0$, we can write $p_i = (\epsilon_i, \vec{p}_i)$, and since $\epsilon_i^2 - \vec{p}_i^2 = 0$ this gives $p_i = (|\vec{p}_i|, \vec{p}_i)$. In the COM system, we can write $\vec{p}_1 = -\vec{p}_2 \equiv \vec{p}_i$ and $\vec{p}_3 = -\vec{p}_4 \equiv \vec{p}_f$. And since $p_1 + p_2 = p_3 + p_4$, we get $|\vec{p}_i| = |\vec{p}_f| \equiv p$. Now we can write:

$$p_1 = (p, \vec{p}_i) \quad p_2 = (p, -\vec{p}_i) \quad p_3 = (p, \vec{p}_f) \quad p_4 = (p, -\vec{p}_f) \quad (57)$$

$$s = 4p^2 \quad t = u = (\vec{p}_i - \vec{p}_f)^2 = 2p^2 - 2\vec{p}_i \cdot \vec{p}_f = 2p^2(1 - \cos\theta) \quad (58)$$

Where θ is the collision angle. So, we get:

$$i\mathcal{M}_{abcd} = -i6\lambda(\delta_{ab}\delta_{cd} + \delta_{ac}\delta_{bd} + \delta_{ad}\delta_{bc}) - i2\lambda\mu^2 \left[\frac{\delta_{ab}\delta_{cd}}{2p^2 - \mu^2 + i\epsilon} + \frac{\delta_{ac}\delta_{bd} + \delta_{ad}\delta_{bc}}{p^2(1 - \cos\theta) - \mu^2 + i\epsilon} \right] \quad (59)$$

3.3 For zero momentum

In the $\vec{p}_i = 0$ case, we can write $p = 0$ or $s = t = u = 0$, giving:

$$\begin{aligned} i\mathcal{M}_{abcd} &= -i6\lambda(\delta_{ab}\delta_{cd} + \delta_{ac}\delta_{bd} + \delta_{ad}\delta_{bc}) - i2\lambda\mu^2 \left[\frac{\delta_{ab}\delta_{cd}}{-\mu^2 + i\epsilon} + \frac{\delta_{ac}\delta_{bd} + \delta_{ad}\delta_{bc}}{-\mu^2 + i\epsilon} \right] = \\ &= -i4\lambda[\delta_{ab}\delta_{cd} + \delta_{ac}\delta_{bd} + \delta_{ad}\delta_{bc}] \end{aligned} \quad (60)$$

In the case of two fields, for instance $a = b \neq c = d$ (any permutation will yield the same result), we get:

$$\mathcal{M} = -4\lambda \quad (61)$$

In the case of all fields being the same $a = b = c = d$, we get:

$$\mathcal{M} = -12\lambda \quad (62)$$

3.4 For small momentum

For small momenta, using:

$$\frac{1}{x - \mu^2} \approx - \left(\frac{1}{\mu^2} + \frac{x}{\mu^4} \right) \quad (63)$$

We get:

$$\begin{aligned} i\mathcal{M}_{abcd} &= -i6\lambda(\delta_{ab}\delta_{cd} + \delta_{ac}\delta_{bd} + \delta_{ad}\delta_{bc}) - \\ &+ i2\lambda\mu^2 \left[\delta_{ab}\delta_{cd} \left(\frac{1}{\mu^2} + \frac{p^2}{\mu^4} \right) + (\delta_{ac}\delta_{bd} + \delta_{ad}\delta_{bc}) \left(\frac{1}{\mu^2} + \frac{p^2(1 - \cos\theta)}{\mu^4} \right) \right] = \\ &= -i4\lambda(\delta_{ab}\delta_{cd} + \delta_{ac}\delta_{bd} + \delta_{ad}\delta_{bc}) + i2\lambda[\delta_{ab}\delta_{cd} + (\delta_{ac}\delta_{bd} + \delta_{ad}\delta_{bc})(1 - \cos\theta)] \frac{p^2}{\mu^2} \end{aligned} \quad (64)$$

For $a = b \neq c = d$:

$$\mathcal{M} = -4\lambda + 2\lambda \left(\frac{p}{\mu} \right)^2 \quad (65)$$

For $a = c \neq b = d$ or $a = d \neq b = c$, we get:

$$\mathcal{M} = -4\lambda + 2\lambda(1 - \cos\theta) \left(\frac{p}{\mu} \right)^2 \quad (66)$$

And for $a = b = c = d$, we get both:

$$\mathcal{M} = -4\lambda + 2\lambda(3 - 2\cos\theta) \left(\frac{p}{\mu} \right)^2 \quad (67)$$

The amplitude is not zero, since there is a mixing of the fields, even in small momenta.

3.5 Symmetry breaking term

3.5.1 Global symmetry

This term breaks the same $SO(N)$ symmetry that was broken when close to the minima, so we get the $SO(N-1)$ rotating all scalar fields except Ψ_N , only now it is broken explicitly in the Lagrangian, and not just for small momenta.

3.5.2 The classical potential minimum

We can write the classic potential:

$$V(\Phi^i) = -\frac{1}{2}\mu^2\Phi_i^2 + \frac{\lambda}{4}[\Phi_i^2]^2 + \alpha\Phi_N \quad (68)$$

Taking the derivative with respect to Φ_N is now different, giving:

$$\frac{\partial V}{\partial \Phi_N} = -\mu^2\Phi_N + \lambda\Phi_i^2\Phi_N + \alpha \quad (69)$$

Taking this to be zero we get the minima for all fields:

$$\Phi_N(\mu^2 - \lambda\Phi_i^2) = \alpha \quad (70)$$

This is still an $N-1$ dimensional manifold, and therefore this doesn't give an additional symmetry breaking.

3.5.3 The mass of the fields around the minima in leading order

Let us choose a minimum from the manifold. We will choose:

$$\Phi_i = \left(0, \dots, \frac{\alpha}{\mu^2}\right) \quad (71)$$

* The time is over and I finally got here. I know this choice is illegal, since it doesn't solve the above equation. For some reason I forgot that Φ_i inside the parantheses includes Φ_N as well when I solved this. I would just solve it in perturbation theory as suggested if I had time, then plug it in. I suppose the mass I would receive is different, maybe zero, but the calculations are the same.

We can now take the same expansion around the minimum we would have taken in the regular SSB case:

$$\Phi_i = (\Pi_k, v + \sigma) \quad v = \frac{\alpha}{\mu^2} \quad (72)$$

And plug it into the Lagrangian. The mass term will give the fields their original mass:

$$\frac{1}{2}\mu^2\Phi_i^2 = \frac{1}{2}\mu^2\Pi_i^2 + \frac{1}{2}\mu^2(v+\sigma)^2 = \frac{1}{2}\mu^2\Pi_i^2 + \frac{1}{2}\mu^2\sigma^2 + \mu^2v\sigma + \frac{1}{2}v^2 \quad (73)$$

While the part from the interaction term that has the fields interacting with v reduces their mass:

$$-\frac{\lambda}{4}[\Phi^2](v+\sigma)^2 = -\frac{\lambda}{4}[\Phi^2]v^2 + \dots = -\frac{\lambda}{4}[\Phi^2]\frac{\alpha^2}{\mu^4} + \dots \quad (74)$$

So, depending on α , the fields might still have mass. Their mass term is:

$$\frac{1}{2}\left(\mu^2 - \frac{\alpha^2}{2\mu^4}\right)\Phi_i^2 \quad (75)$$